

Graphs, Surfaces and Homology

Third Edition

Homology theory is a powerful algebraic tool that is at the centre of current research in topology and its applications. This accessible textbook will appeal to students interested in the application of algebra to geometrical problems, specifically the study of surfaces (such as sphere, torus, Möbius band, Klein bottle). In this introduction to simplicial homology – the most easily digested version of homology theory – the author studies interesting geometrical problems, such as the structure of two-dimensional surfaces and the embedding of graphs in surfaces, using the minimum of algebraic machinery and including a version of Lefschetz duality.

Assuming very little mathematical knowledge, the book provides a complete account of the algebra needed (abelian groups and presentations), and the development of the material is always carefully explained with proofs of all the essential results given in full detail. Numerous examples and exercises are also included, making this an ideal text for undergraduate courses or for self-study.

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For Rachel

